Note of Fast Accurate Eigenvalue Computations Using The Cholesky Factorization

1 LU

See Ref. [6]

Assume $H = DAD$,

$$LL^T = H$$

Theorem 2.3:

Let $G = BD \in \mathbb{R}^{n\times n}$ be non-singular and let $\hat{G} = (B + \Delta B)D$ where $\|\Delta B\| \leq \sigma_n(B)$.

$$\left|\frac{\sigma_i(\hat{G}) - \sigma_i(G)}{\sigma_i(G)}\right| \leq \frac{\|\Delta B\|}{\sigma_n(B)}, \quad i = 1, \ldots, n.$$

Co

$$G = L\tilde{D}^{-1}\tilde{D}$$

$$\tilde{G} = (L\tilde{D}^{-1} - \Delta\tilde{D}^{-1})\tilde{D} = L_{(1)}\tilde{D}^{-1}\tilde{D}$$

$$\left\|L\tilde{D}^{-1}\right\| \left|\frac{\sigma_i(L_{(1)}) - \sigma_i(L)}{\sigma_i(L)}\right| \leq \frac{\left\|\Delta\tilde{D}^{-1}\right\|}{\sigma_n(L\tilde{D}^{-1})}$$

2 Stronger version

For graded Assume $\sigma_{k-1}(L) \gg \sigma_k(L) \gg \sigma_{k+1}(L)$, can we prove eigenvector is close to id?

Or, if $\sigma_{k-j}(A) \gg \sigma_{k+l}(A)$, $\forall j, l \geq 1$, can we prove the singular space is seperated?

3 Eigenvalue bounds

3.1 Additive perturbations

3.1.1 Bauer-Fike-type bounds for diagonalizable matrices.

For $A = X\Lambda X^{-1}$(with eigenvalues $\{\lambda_i\}$), and perturbation $A + E$ (with an eigenvalue $\hat{\lambda}$), then:

$$\min_i \left|\lambda_i - \hat{\lambda}\right| \leq \kappa(X)\,\|E\|\,. \tag{1}$$

where $\kappa(X)$ is condition number of X:

$$\kappa(X) = \|X\| \left\|X^{-1}\right\| \tag{2}$$

For relative bound (Theorem 2.1 and 2.3 in Ref. [2])

$$\min_i \frac{\left|\lambda_i - \hat{\lambda}\right|}{|\lambda_i|} \leq \kappa(X) \left\|A^{-1}E\right\|. \tag{3}$$

$$\min_i \frac{\left|\lambda_i - \hat{\lambda}\right|}{|\lambda_i|} \leq \kappa(X) \left\|A_1^{-1}EA_2^{-1}\right\|. \tag{4}$$

where $A = A_1A_2 = A_2A_1$.

3.1.2 Whoffman-Wielandt theorem

For $A = X\Lambda X^{-1}$(with eigenvalues $\{\lambda_i\}$), and perturbation $A+E = \hat{X}\hat{\Lambda}\hat{X}^{-1}$ (with eigenvalues $\left\{\hat{\lambda}\right\}$), then:

$$\sqrt{\sum_{i=1}^{n} \left|\lambda_i - \hat{\lambda}_{\tau(i)}\right|^2} \leq \kappa(\hat{X})\kappa(X) \|E\|_F$$

for some pertmutation τ. And (Theorem 2.6 in Ref. [2])

$$\sqrt{\sum_{i=1}^{n} \left(\frac{\left|\lambda_i - \hat{\lambda}_{\tau(i)}\right|}{|\lambda_i|}\right)^2} \leq \kappa(\hat{X})\kappa(X) \left\|A^{-1}E\right\|_F$$

For Hermitian matrix and sort eigenvalues in descending order, then

$$\sqrt{\sum_{i=1}^{n} \left|\lambda_i - \hat{\lambda}_i\right|^2} \leq \|E\|_F$$

(Theorem 2.7 in Ref. [2]) If, in addition, A and $A + E$ are Hermitian positive-definite, then

$$\sqrt{\sum_{i=1}^{n} \frac{\left|\lambda_i - \hat{\lambda}_{\tau(i)}\right|^2}{\lambda_i\hat{\lambda}_i}} \leq \left\|A^{-1/2}EA^{-1/2}\left(I + A^{-1/2}EA^{-1/2}\right)^{-1/2}\right\|_F \tag{5}$$

3.1.3 Weyl-type bounds

For Hermitian matrix A (with eigenvalues $\{\lambda_i\}$) and $A + E$ (with eigenvalues $\left\{\hat{\lambda}\right\}$).

$$\max_i \left|\lambda_i - \hat{\lambda}_i\right| \leq \|E\|.$$

(Theorem 2.8 in Ref. [2]) If further, A is positive-definite, then

$$\max_i \frac{\left|\lambda_i - \hat{\lambda}_i\right|}{|\lambda_i|} \leq \left\|A^{-1/2}EA^{-1/2}\right\|.$$

(Theorem 2.9 in Ref. [2]) If further, $A+E$ is also positive-definite, then

$$\max_i \frac{\left|\lambda_i - \hat{\lambda}_i\right|}{\sqrt{\lambda_i \hat{\lambda}_i}} \leq \left\| A^{-1/2} E A^{-1/2} \left(I + A^{-1/2} E A^{-1/2}\right)^{-1/2} \right\|. \tag{6}$$

(Theorem 2.11 in Ref. [2]) A and $A+E$ be Hermitian. H is the positive-semidefinite polar factor of A and for some $0 < \epsilon < 1$

$$|x^* E x| \leq \epsilon x^* H x \quad \forall x$$

then

$$\left|\lambda_i - \hat{\lambda}_i\right| \leq \epsilon \left|\lambda_i\right|.$$

(Corollary 2.13 in Ref. [2]) For $A = D^* M D$ Hermitian positive-definite and $A+E = D^*(M+F)D$ Hermitian, D nonsingular, then

$$\max_i \frac{\left|\lambda_i - \hat{\lambda}_i\right|}{|\lambda_i|} \leq \left\| M^{-1/2} F M^{-1/2} \right\|.$$

similarily, the Eq.(5) and Eq.(6).

(Theorem 2.19 in Ref. [2]) Let $A = DMD$ and $A+E = D(M+F)D$ be real symmetric and D a real nonsingular diagonal matrix. Among all eigenvalues of principal submatrices of M, let μ be the smallest in magnitude. If $\|F\| < |\mu|$ then

$$-\|F\| \frac{2|\mu| - \|F\|}{|\mu|^2} \leq \frac{\hat{\lambda}_i - \lambda_i}{\lambda_i} \leq \|F\| \frac{2|\mu| - \|F\|}{(|\mu| - \|F\|)^2} \tag{7}$$

3.1.4 Ritz values

Let A be a Hermitian matrix of order n and $Q \in \mathbb{C}^{n\times m}$, $Q^*Q = I$. Then $W = Q^*AQ \in \mathbb{C}^{m\times m}$, whose eigenvalues $\left\{\hat{\lambda}_j\right\}$ are called Ritz values of A.

Define $R = AQ - QW$, then one can always find m eigenvalues $\left\{\lambda_{\tau(j)}\right\}$ of A, such that

$$\max_j \left|\lambda_{\tau(j)} - \hat{\lambda}_j\right| \leq \|R\| \tag{8}$$

3.2 Additive pertubations fo singular values

Assume $B \in \mathbb{C}^{m\times n}$, $m \geq n$.

(Theorem 3.2 in Ref. [2]) If $\text{range}(B+F) \subset \text{range}(B)$, then

$$|\sigma_i - \hat{\sigma}_i| \leq \sigma_i \left\|B^+ F\right\|, \quad 1 \leq i \leq n.$$

If $\text{range}((B+F)^*) \subset \text{range}(B^*)$, (or weaker, Corollary 3.3, B is full column rank), then

$$|\sigma_i - \hat{\sigma}_i| \leq \sigma_i \left\|F B^+\right\|, \quad 1 \leq i \leq n.$$

(Theorem 3.9 in Ref. [2]) Let $B = D_l C D_r$ and $B+F = D_l(C+G)D_r$ be real symmetric, where D_l and D_r are real nonsingular diagonal matrices. Among the singular values of all square submatrices of B (?? maybe typo here), let θ be the smallest one. If $\|G\| < \theta$, then

$$-\|G\| \frac{2\theta - \|G\|}{\theta^2} \leq \frac{\hat{\sigma}_i - \sigma_i}{\sigma_i} \leq \|G\| \frac{2\theta - \|G\|}{(\theta - \|G\|)^2} \quad \forall i \tag{9}$$

3.3 Graded Matrices perturbation

(Theorem 5.4 in Ref. [3]) Let $A = S^*HS$ and $\tilde{A} = S^*\tilde{H}S$ be two $n \times n$ nonnegative definite Hermitian matrices with eigenvalues ordered descendingly, let $\Delta H = \tilde{H} - H$. If $\|H^{-1}\| \, \|\Delta H\| < 1$, then

$$\max_i \chi(\lambda_i, \tilde{\lambda}_i) \leq \frac{\|H^{-1}\| \, \|\Delta H\|}{\sqrt{1 - \|H^{-1}\| \, \|\Delta H\|}}$$

$$\sqrt{\sum_{i=1}^{n} \left(\chi(\lambda_i, \tilde{\lambda}_i)\right)^2} \leq \frac{\|H^{-1}\| \, \|\Delta H\|_F}{\sqrt{1 - \|H^{-1}\| \, \|\Delta H\|}}$$

where (also define $0/0 = 0$)

$$\chi(a, \tilde{a}) = \frac{|a - \tilde{a}|}{\sqrt{a\tilde{a}}}.$$

(Theorem 5.5 in Ref. [3]) Let $B = GS$ and $\tilde{B} = \tilde{G}S$ be two $n \times n$ matrices with singular values ordered descendingly, G and $\tilde{G}$ are nonsingular, let $\Delta G = \tilde{G} - G$. If $\|\Delta G\| \, \|G^{-1}\| < 1$, then

$$\max_i \chi(\sigma_i, \tilde{\sigma}_i) \leq \left(1 + \frac{1}{1 - \|G^{-1}\| \, \|\Delta G\|}\right) \frac{\|G^{-1}\| \, \|\Delta G\|}{2}$$

$$\sqrt{\sum_{i=1}^{n} \left(\chi(\sigma_i, \tilde{\sigma}_i)\right)^2} \leq \left(1 + \frac{1}{1 - \|G^{-1}\| \, \|\Delta G\|}\right) \frac{\|G^{-1}\| \, \|\Delta G\|_F}{2}.$$

(Theorem 7.5 in Ref. [3]) Under the conditions above, we have

$$\frac{\left|\tilde{\lambda}_j - \lambda_j\right|}{\lambda_j} \leq \left\|H^{-1}\right\| \, \|\Delta H\| \quad \forall j,$$

$$\frac{|\tilde{\sigma}_j - \sigma_j|}{\sigma_j} \leq \left\|G^{-1}\right\| \, \|\Delta G\| \quad \forall j.$$

4 Derivative of eigenvalues and eigenvectors

Derivative of an eigenvalue $\lambda(X)$ and an eigenvector $u(X)$ of matrix X at X_0.

$\cdot^+$ means Moore-Penrose inverse.

TODO:

1. Error bound estimation? Grönwall's inequality?
2. SVD when has non-simple roots?

4.1 Real symmetric case

$X_0 \in \mathbb{R}^{n \times n}$, $X_0 = X_0^T$:

$$X_0 u_0 = \lambda_0 u_0$$

and λ_0 be a simple eigenvalue, then

$$Xu = \lambda u, \quad u^T u = 1, \quad X \in N(X_0)$$

$$\mathrm{d}\lambda = u_0^T(\mathrm{d}X)u_0$$
$$\mathrm{d}u = (\lambda_0 I_n - X_0)^+(\mathrm{d}X)u_0.$$

Note that $X_0 = U_0\Lambda_0 U_0^T$, so

$$\mathrm{d}u = U_0\left(\lambda_0 I_n - \Lambda_0\right)^+ U_0^T(\mathrm{d}X)u_0,$$

which is a specific column of Eq.(11).

4.2 General case

4.2.1 Normalization $u_0^* u = 1$

See [5]

$Z_0 \in \mathbb{C}^{n\times n}$:

$$Z_0 u_0 = \lambda_0 u_0$$

and λ_0 be simple eigenvalue, then

$$Zu = \lambda u, \quad u_0^* u = 1, \quad Z \in N(Z_0)$$

$$\mathrm{d}\lambda = \frac{v_0^*(\mathrm{d}Z)u_0}{v_0^* u_0}$$

$$\mathrm{d}u = (\lambda_0 I - Z_0)^+ \left(I - \frac{u_0 v_0^*}{v_0^* u_0}\right)(\mathrm{d}Z)u_0 \tag{10}$$

where $Z_0^* v_0 = \bar{\lambda}_0 v_0$, i.e. v_0^* is the left eigenvector of Z_0 corresponding to λ_0.

4.2.2 Normalization diagC=0

See [1]

Assume: $A \in \mathbb{R}^{n\times n}$, $\lambda_i \neq \lambda_j \ \forall i \neq j$. $AU = U\Lambda$.

$$\mathrm{d}\Lambda = I \circ \left(U^{-1}\mathrm{d}A\, U\right).$$

Normalization: $\mathrm{diag}\left(U^{-1}\mathrm{d}U\right) = 0$:

$$\mathrm{d}U = U\left(F \circ \left(U^{-1}\mathrm{d}A\, U\right)\right) \tag{11}$$

where $F_{ij} = (\lambda_j - \lambda_i)^{-1}$ for $i \neq j$, and zero otherwise.

In general, Eq.(10) is different from Eq.(11). Eq.(11) seems has problem with complex conjugate roots. The authur only verified real eigenvalue cases. Also it needs extra normalization step to get accurate result.

The underlying difficulty is that $|z|$ or more specifically $\bar{z}$ is not an analytical function. i.e. not able to find a constant C (independent from $\mathrm{d}z$) in $\mathrm{d}f = C\mathrm{d}z$. This pose a problem when derive the derivative of normalized vector. See Sec.7 for a potiential solution.

5 Derivative of SVD

See [1].

Assume $A \in \mathbb{R}^{m\times n}$,

$$A = USV^T,$$

then

$$\mathrm{d}S = I \circ \left(U^T \mathrm{d}A\, V\right).$$

Assume $m \le n$.

Defined $\mathrm{d}C = U^{-1}\mathrm{d}U$, $\mathrm{d}D = V^{-1}\mathrm{d}V$ (note $\mathrm{d}C$ and $\mathrm{d}D$ are anti-symmetric), and partition

$$S = \begin{bmatrix} S_1 & 0 \end{bmatrix}, \quad \mathrm{d}S = \begin{bmatrix} \mathrm{d}S_1 & 0 \end{bmatrix}, \quad \mathrm{d}D = \begin{bmatrix} \mathrm{d}D_1 & -\mathrm{d}D_2 \\ \mathrm{d}D_2^T & \mathrm{d}D_3 \end{bmatrix}$$

$$U^T \mathrm{d}A\, V = \begin{bmatrix} \mathrm{d}P_1 & \mathrm{d}P_2 \end{bmatrix}$$

then

$$\mathrm{d}D_2 = S_1^{-1}\mathrm{d}P_2$$
$$\mathrm{d}C = F \circ \left(\mathrm{d}P_1 S_1 + S_1 \mathrm{d}P_1^T\right)$$
$$\mathrm{d}D_1 = F \circ \left(S_1 \mathrm{d}P_1 + \mathrm{d}P_1^T S_1\right)$$

where $F_{ij} = \left(s_j^2 - s_i^2\right)^{-1}$ for $i \ne j$ and zero otherwise.

$$\mathrm{d}U = U\left(F \circ \left(\mathrm{d}P_1\, S_1 + S_1 \mathrm{d}P_1^T\right)\right)$$

$$\begin{bmatrix} \mathrm{d}V_1 \\ \mathrm{d}V_2 \end{bmatrix} = V \begin{bmatrix} \mathrm{d}D_1 \\ \mathrm{d}D_2^T \end{bmatrix}.$$

Assume $m \ge n$, $A = B^T$, $B = U_B S_B V_B^T$, (economical format, i.e. $U_B \in \mathbb{R}^{m\times n}$, $S_B \in \mathbb{R}^{n\times n}$, $V_B \in \mathbb{R}^{n\times n}$), then

$$A = USV^T = B^T = V_B S_B U_B^T$$
$$V_B = U, \quad U_B = V$$

Thus, using orginal symbols, we have

$$\mathrm{d}S = I \circ \left(U^T \mathrm{d}A\, V\right).$$

$$\begin{bmatrix} \mathrm{d}U_1 \\ \mathrm{d}U_2 \end{bmatrix} = U \begin{bmatrix} \mathrm{d}C_1 \\ \mathrm{d}C_2^T \end{bmatrix}.$$
$$\mathrm{d}D = F \circ \left(\mathrm{d}P_1^T S_1 + S_1 \mathrm{d}P_1\right)$$
$$\mathrm{d}C_1 = F \circ \left(S_1 \mathrm{d}P_1^T + \mathrm{d}P_1 S_1\right)$$

$$\mathrm{d}C_2 = S_1^{-1}\mathrm{d}P_2^T$$
$$U^T \mathrm{d}A\, V = \begin{bmatrix} \mathrm{d}P_1 \\ \mathrm{d}P_2 \end{bmatrix}$$

draft——————————————————— For economical format, $A \in \mathbb{R}^{m\times n}$, $U \in \mathbb{R}^{m\times p}$, $S \in \mathbb{R}^{p\times p}$, $V \in \mathbb{R}^{p\times n}$, $p = \min\{m, n\}$

$$A = USV^T$$

Define $\mathrm{d}C = U^T\mathrm{d}U$, $\mathrm{d}D = V^T\mathrm{d}V$, then $\mathrm{d}C$ and $\mathrm{d}D$ are anti-symmetric ($\mathrm{d}C + \mathrm{d}C^T = 0$, $\mathrm{d}D + \mathrm{d}D^T = 0$), and

$$\mathrm{d}A = \mathrm{d}U\, SV^T + U\mathrm{d}S\, V^T + US\mathrm{d}V^T$$
$$U^T\mathrm{d}A\, V = \mathrm{d}C\, S + \mathrm{d}S + S\mathrm{d}D^T$$

Then

$$\mathrm{d}S = I \circ \left(U^T\mathrm{d}A\, V\right)$$

Define and solve

$$\mathrm{d}P \triangleq U^T\mathrm{d}A\, V$$
$$\mathrm{d}P = \mathrm{d}C\, S + \mathrm{d}S - S\mathrm{d}D$$
$$\mathrm{d}P^T = -S\mathrm{d}C + \mathrm{d}S + \mathrm{d}D\, S$$
$$\mathrm{d}P\, S + S\mathrm{d}P^T = \mathrm{d}C\, S^2 - S^2\mathrm{d}C + 2\mathrm{d}S\, S$$
$$S\mathrm{d}P + \mathrm{d}P^T\, S = \mathrm{d}D\, S^2 - S^2\mathrm{d}D + 2\mathrm{d}S\, S$$

Due to anti-symmetric

$$\mathrm{d}C = F \circ \left(\mathrm{d}P\, S + S\mathrm{d}P^T\right)$$
$$\mathrm{d}D = F \circ \left(S\mathrm{d}P + \mathrm{d}P^T\, S\right)$$

? how to get $\mathrm{d}U$? What is UU^T?
Mathematical equivlant form, assume AA^T is rank defect, we still have

$$US^2U^T = AA^T$$

Then

$$\mathrm{d}S^2 = I \circ \left(U^{-1}\mathrm{d}(AA^T)\, U\right).$$

6 Derivative of Cholesky decomposition

Suppose $\Sigma = \Sigma^T \in \mathbb{R}^{n\times n}$,

$$LL^T = \Sigma$$

with perturbation $\mathrm{d}\Sigma = \mathrm{d}\Sigma^T$, then

$$\mathrm{d}L = L\,\Phi(L^{-1}\mathrm{d}\Sigma\left(L^{-1}\right)^T),$$

where the Φ operator extract the lower-triangular part of a matrix and halves its diagonal:

$$\Phi_{ij}(A) = \begin{cases} A_{ij} & i > j \\ \frac{1}{2}A_{ii} & i = j \\ 0 & i < j \end{cases}.$$

7 Derivative of normalize operator

Suppose $x \in \mathbb{R}^{n\times 1}$

$$y = \frac{x}{\|x\|},$$

then

$$\mathrm{d}y = \left(I - \frac{xx^T}{\|x\|^2}\right)\frac{\mathrm{d}x}{\|x\|}.$$

For vector $z \in \mathbb{C}^{n\times 1}$.

A convention used in autograd (Ref. [4]):

$$f(z) = u(\mathrm{Re}(z), \mathrm{Im}(z)) + \mathrm{i}vu(\mathrm{Re}(z), \mathrm{Im}(z)),$$

the derivative is defined by

$$\frac{\mathrm{d}f(z)}{\mathrm{d}z} = \frac{\partial u}{\partial x}(\mathrm{Re}(z), \mathrm{Im}(z)) - \mathrm{i}\frac{\partial u}{\partial y}(\mathrm{Re}(z), \mathrm{Im}(z)).$$

Under this definition

$$\frac{\mathrm{d}}{\mathrm{d}z}\frac{z}{\sqrt{z^*z}} = \frac{1}{\sqrt{z^*z}} - \frac{z}{2}\left(z^*z\right)^{-\frac{3}{2}}\frac{\mathrm{d}}{\mathrm{d}z}\left(z^*z\right)$$

where

$$u(x, y) = x^2 + y^2$$

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(z^*z\right) = 2\mathrm{Re}(z^T) - 2\mathrm{i}\mathrm{Im}(z^T) = 2z^*$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}z}\frac{z}{\sqrt{z^*z}} = \frac{1}{\|z\|}\left(I - \frac{zz^*}{\|z\|^2}\right)$$

8 Grönwall's inequality

If

$$u'(t) \le \beta(t)u(t), \quad \forall t \in (a, b)$$

then

$$u(t) \le u(a)\exp(\int_a^t \beta(s)\,\mathrm{d}s)$$

9 Taylor's theorem

For Taylor series

$$f(x) = \sum_{j=0}^{k}\frac{f^{(j)}(a)}{j!}(x-a)^j + R_k(x).$$

Lagrange form of the remainder, $\exists \xi_L \in [a, x]$

$$R_k(x) = \frac{f^{(k+1)}(\xi_L)}{(k+1)!}(x-a)^{k+1}.$$

Cauchy form of the remainder, $\exists \xi_C \in [a,x]$

$$R_k(x) = \frac{f^{(k+1)}(\xi_C)}{k!}(x-\xi_C)^k(x-a).$$

General form: for $G \in C^1([a,x])$, and $G'(t) \neq 0, \forall t \in (a,x)$, $\exists \xi \in [a,x]$

$$R_k(x) = \frac{f^{(k+1)}(\xi)}{k!}(x-\xi)^k \frac{G(x)-G(a)}{G'(\xi)}.$$

Integral form of the remainder: $f^{(k)}$ be absolutely continuous on $[a,x]$

$$R_k(x) = \int_a^x \frac{f^{(k+1)}(t)}{k!}(x-t)^k \mathrm{d}t.$$

Remainder in complex analysis:

$$R_k(z) = \frac{(z-a)^{k+1}}{2\pi \mathrm{i}} \int_{\partial\Omega} \frac{f(s)\,\mathrm{d}s}{(s-a)^{k+1}(s-z)}$$

with bounds for $z \in \Omega = \{s \in \mathbb{C} | \, |s-a| \leq r\}$

$$\left|f^{(k)}(z)\right| \leq \frac{k! M_r}{r^k}, \quad M_r = \max_{|s-a|=r} |f(s)|$$

$$R_k(z) \leq \frac{\beta^{k+1} M_r}{1-\beta}, \quad \beta = \frac{|z-a|}{r} < 1$$

10 Homotopy Transform

$t \in [0,1]$, $(1-t)A + (1-t)tC + tB$, not work.

$(1-t)A + (1-t)tC(t) + tB$ might work, but no bound for $C(t)$.

11 Dynamics of eigenvector

It is

$$\mathrm{d}\mathrm{diag}\Lambda = \mathrm{diag}\left(U^{-1}\mathrm{d}A\,U\right).$$
$$\mathrm{d}U = U\left(F \circ \left(U^{-1}\mathrm{d}A\,U\right)\right)$$

12 Rude approximation asymptotic

Due to

$$\mathrm{d}u = U_0\left(\lambda_0 I_n - \Lambda_0\right)^+ U_0^T(\mathrm{d}X)u_0,$$

the graded matrix expansion

$$\mathrm{d}u = \left(\lambda_0 I_n - \Lambda_0\right)^+ (A - \mathrm{diag}(A))u_0$$

Or

$$\mathrm{d}U = F \circ (A - \mathrm{diag}(A))$$

where $F_{ij} = (\lambda_j - \lambda_i)^{-1}$ for $i \neq j$, and zero otherwise.

In the exponential cases

$$|a_{kl}| \leq C_0 \rho^{-k}, \quad \forall k, l \geq 0$$
$$|K_{jk}| \leq C_0 L \rho^{-j-k}.$$

Approx

$$(\lambda_j - \lambda_i)^{-1} \sim (C_0 L)^{-1} \left(\rho^{-2j} - \rho^{-2i}\right)^{-1}$$

$$\begin{aligned}
(F \circ (A - \mathrm{diag}(A)))_{ij} &\sim (C_0 L)^{-1} \left(\rho^{-2j} - \rho^{-2i}\right)^{-1} C_0 L \rho^{-i-j} \\
&= \left(\rho^{-2j} - \rho^{-2i}\right)^{-1} \rho^{-i-j} \\
&= \begin{cases} \rho^{j-i} & j \ll i \\ -\rho^{i-j} & i \ll j \end{cases}
\end{aligned}$$

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